Hochschule Fulda
University of Applied Sciences

Stock price Prediction & Portfolio Optimization

Course

Application of Data Science in Finance (AI5247)

Department

Applied Computer Science

Submitted to

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1. Introduction

1.1 Background and Problem Statement

Today's financial world is riddled with extreme volatility and high complexity, making traditional methods used for analysis and prediction less effective. The inadequacy of these methods to manage risk and find predictive signals has created a need for better data-driven methods.

This project tries to solve this problem by using various quantitative approaches and data science techniques. The goal is to get above the simple historical analysis and give diverse methods for getting predictions and better portfolio optimisation.

1.2 Project Objectives

To achieve better returns and market insight, this study focuses on three primary objectives, which collectively define the scope of our project:

1. Predictive Modelling: To accurately forecast short-term stock returns for a set of companies by constructing and evaluating machine learning models, specifically using Bayesian Networks to understand market dynamics based on technical indicators.
2. Feature Selection: Here, we identify and validate the most influential financial indicators that drive stock performance. This is done by employing the Genetic Algorithm (GA) with OLS regression to find the optimal predictive factor combinations.
3. Portfolio Optimisation: Using modern computational methods, including the PyPortfolioOpt library, to construct investment portfolios that significantly outperform traditional models. The goal is to determine the optimal strategy to balance between maximising returns and minimising risk, measured using appropriate performance metrics.

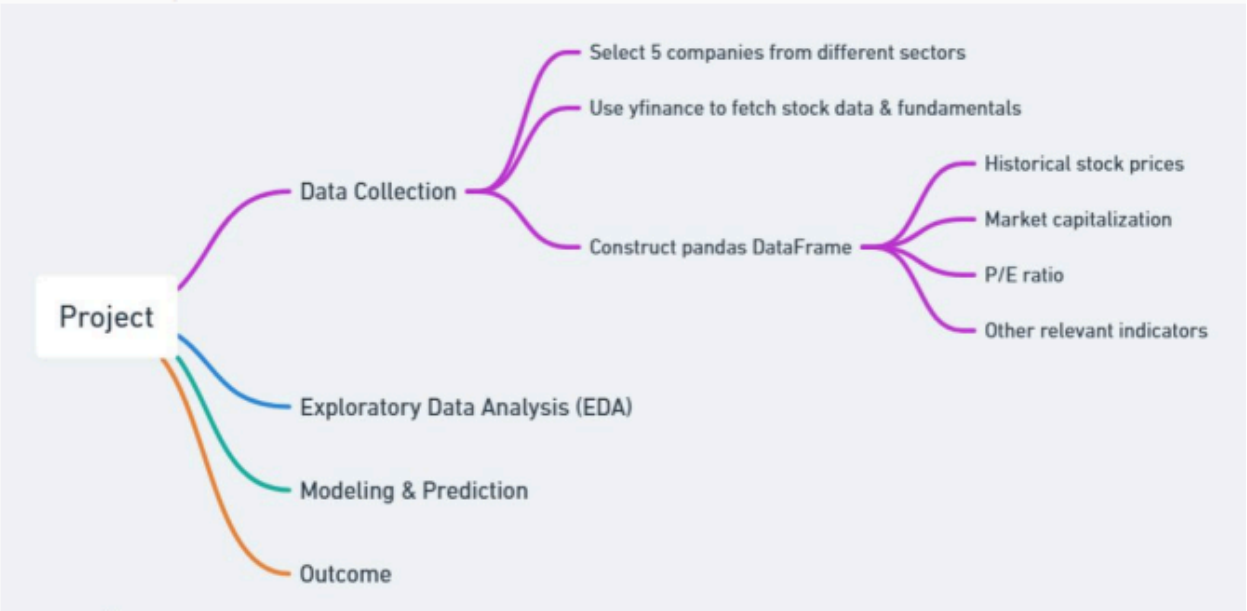
1.3 Project Scope and Report Structure

The analysis is conducted on a portfolio of five major U.S. companies from different sectors: Apple (AAPL), JPMorgan Chase & Co. (JPM), Johnson & Johnson (JNJ), Exxon Mobil Corporation (XOM), and Procter & Gamble (PG). Historical financial data, spanning from 2017 to 2025, is collected using the yfinance Python library to form the training set, with the period from January 2025 to September 2025 reserved for testing only.

The report proceeds with details of the Data Acquisition and Feature Engineering process, including the calculation of financially relevant indicators. Then, it covers the Exploratory Data Analysis (EDA) and presents the Implementation and Results of the predictive models, feature selection and evaluates the Portfolio Optimisation strategies using tools like quantstats.

1.4 Project overview and Dataset Understanding

Project Outline:



Diverse Data Collection

Our analysis is built upon a dataset from five major companies across varied sectors, ensuring a comprehensive market perspective. This diversity mitigates sector-specific biases and provides an overall view of market dynamics.

● AAPL (Technology): Apple Inc. | JPM (Financial): JPMorgan Chase & Co | JNJ (Healthcare): Johnson & Johnson | XOM (Energy): Exxon Mobil Corporation PG (Consumer Goods): Procter & Gamble

Comprehensive Dataset Parameters

Each company's profile includes an extensive array of financial data points, enabling deep analysis and precise model training. This includes both historical performance and fundamental health indicators.

Historical Performance

- Daily Prices (Open, High, Low, Close)
- Trading Volumes

Technical Indicators

- RSI (Relative Strength Index)
- Moving Averages

Fundamental Metrics

- P/E Ratios
- Market Capitalization
- Beta Values
- Dividend Yields

Exploratory Data Analysis (EDA)

Before modelling, we conducted a thorough EDA to understand the underlying patterns and characteristics of our financial data. This step is crucial for identifying trends, anomalies, and informing feature engineering.

Date	Open	High	Low	Close	Volume	Dividends	Stock Splits
2023-09-19 00:00:00-04:00	146.252833	146.547715	144.340850	146.110153	4704200	0.0	0.0
2023-09-20 00:00:00-04:00	146.271870	147.232615	145.881859	145.986496	5446400	0.0	0.0
2023-09-21 00:00:00-04:00	145.967463	146.214795	144.654765	144.721344	5096400	0.0	0.0
2023-09-22 00:00:00-04:00	145.006740	145.862849	143.998430	144.188675	6327000	0.0	0.0
2023-09-25 00:00:00-04:00	143.808156	143.903285	142.704731	143.313522	5527000	0.0	0.0
...	...	...	...	...	...	...	...
2024-02-05 00:00:00-05:00	152.363163	153.162693	151.775557	152.401703	6931900	0.0	0.0
2024-02-06 00:00:00-05:00	152.498030	153.230138	152.074182	153.124176	6185400	0.0	0.0
2024-02-07 00:00:00-05:00	153.259019	153.962215	152.912235	153.278275	7209300	0.0	0.0
2024-02-08 00:00:00-05:00	152.970017	153.259006	151.977832	152.815887	6121600	0.0	0.0
2024-02-09 00:00:00-05:00	152.449875	152.526939	151.197613	151.640717	5978000	0.0	0.0

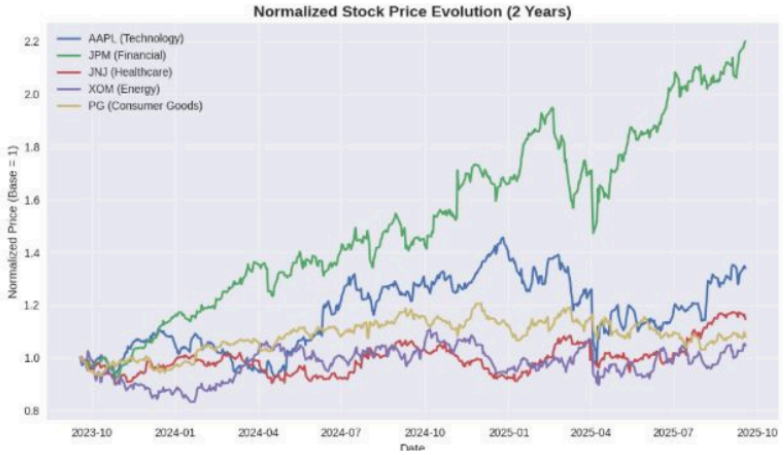
Enhancing/ Modifying the Dataset

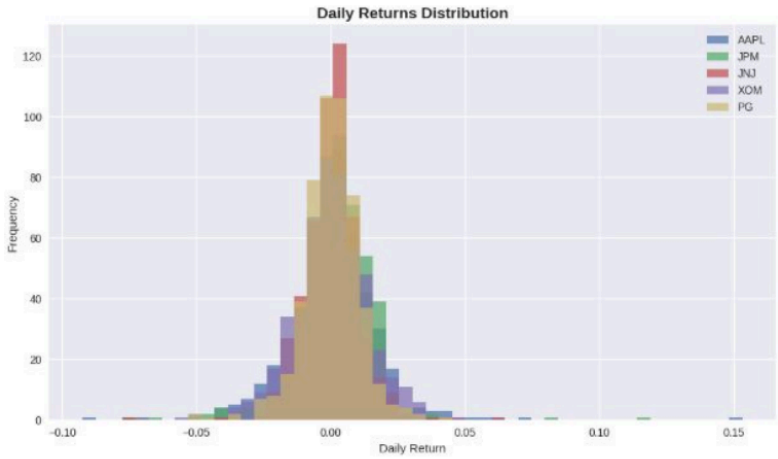
The data set did not contain enough parameters we required to apply models and , so we introduced more parameter ins the dataset

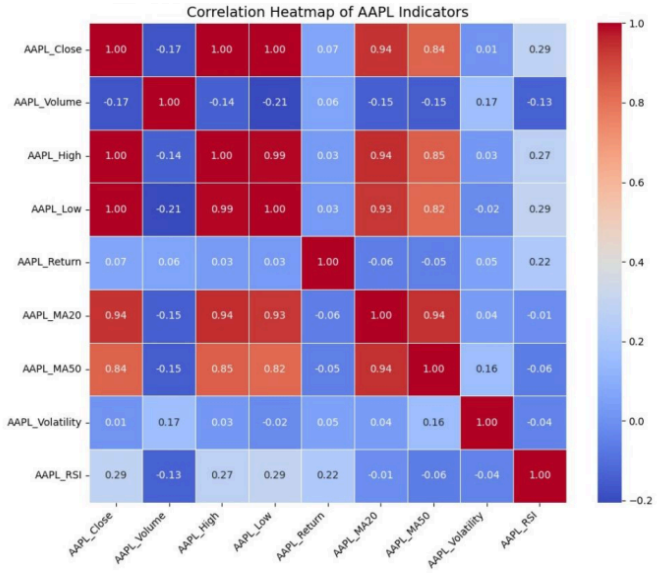
```
Data columns (total 65 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   Date                  1987 non-null   object
 1   Open                  1987 non-null   float64
 2   High                  1987 non-null   float64
 3   Low                   1987 non-null   float64
 4   Close                 1987 non-null   float64
 5   Volume                1987 non-null   int64
 6   Dividends             1987 non-null   float64
 7   Stock Splits          1987 non-null   float64
 8   Return                1987 non-null   float64
 9   Log_Return            1987 non-null   float64
 10  Price_Range           1987 non-null   float64
 11  Volume_Price          1987 non-null   float64
 12  MA_20                 1987 non-null   float64
 13  MA_50                 1987 non-null   float64
 14  MA_200                1987 non-null   float64
 15  MA_Signal_20_50       1987 non-null   int64
 16  MA_Signal_50_200      1987 non-null   int64
 17  EMA_12                1987 non-null   float64
 18  EMA_26                1987 non-null   float64
 19  MACD                  1987 non-null   float64
 20  MACD_Signal           1987 non-null   float64
 21  MACD_Histogram        1987 non-null   float64
 22  RSI                   1987 non-null   float64
 23  BB_Middle             1987 non-null   float64
 24  BB_Upper              1987 non-null   float64
 25  BB_Lower              1987 non-null   float64
 26  BB_Width              1987 non-null   float64
 27  BB_Position           1987 non-null   float64
 28  Volatility_20         1987 non-null   float64
 29  Volatility_50         1987 non-null   float64
 30  GARCH_Vol             1987 non-null   float64
 31  Volume_MA             1987 non-null   float64
 32  Volume_Ratio          1987 non-null   float64
 33  OBV                   1987 non-null   float64
```

```
 34  Momentum_5            1987 non-null   float64
 35  Momentum_10           1987 non-null   float64
 36  Momentum_20           1987 non-null   float64
 37  Support               1987 non-null   float64
 38  Resistance            1987 non-null   float64
 39  Support_Distance      1987 non-null   float64
 40  Resistance_Distance   1987 non-null   float64
 41  Cumulative_Return     1987 non-null   float64
 42  Drawdown              1987 non-null   float64
 43  Max_Drawdown          1987 non-null   float64
 44  Day_of_Week           1987 non-null   int64
 45  Month                 1987 non-null   int64
 46  Quarter               1987 non-null   int64
 47  Year                  1987 non-null   int64
 48  Price_Position        1987 non-null   float64
 49  Gap                   1987 non-null   float64
 50  PE_Ratio              1987 non-null   float64
 51  Beta                  1987 non-null   float64
 52  Dividend_Yield        1987 non-null   float64
 53  Close_Lag_1           1987 non-null   float64
 54  Return_Lag_1          1987 non-null   float64
 55  Volume_Lag_1          1987 non-null   float64
 56  Close_Lag_2           1987 non-null   float64
 57  Return_Lag_2          1987 non-null   float64
 58  Volume_Lag_2          1987 non-null   float64
 59  Close_Lag_3           1987 non-null   float64
 60  Return_Lag_3          1987 non-null   float64
 61  Volume_Lag_3          1987 non-null   float64
 62  Close_Lag_5           1987 non-null   float64
 63  Return_Lag_5          1987 non-null   float64
 64  Volume_Lag_5          1987 non-null   float64
```

Visualization for understanding:







2. Methodology

2.1 Predictive Modelling with Bayesian Networks (Avishkar Sanjay Potale)

For short-term stock returns prediction, a Bayesian Network (BN) model was implemented for each of the five companies. This probabilistic graphical model was chosen for its ability to represent and analyse the complex dependencies and conditional relationships between various financial indicators. Here, we employed HillSearchClimb with BIC to get appropriate results for each company.

The core objective of this modelling phase was to predict the direction of stock returns. The dataset contained various variables from which 9 were selected, with the target variable Return. The model was trained on historical data spanning from 2017 to 2024 and subsequently tested on out-of-sample data from January 1, 2025, to September 12, 2025. This chronological split ensures that the model's predictive performance is evaluated on unseen data, simulating a real-world forecasting scenario.

The implementation focused on understanding the dependency of the target variable, Return_bin (a binarised version of stock return), on other predictive factors. The network structure visualised for each company illustrates the probabilistic influences that indicators like Momentum, RSI, Return_Lag, etc. have on the final predicted return, providing an interpretable model of financial dynamics.

2.2 Genetic Algorithm with Regression (Avishkar Sanjay Potale)

To identify the most influential financial indicators driving stock performance, a hybrid approach combining Genetic Algorithm (GA) with Ordinary Least Squares (OLS) Regression was employed. This method was designed to search through various combinations of predictive factors to find the optimal subset for explaining stock returns.

The process operates as follows:

1. The Genetic Algorithm generates an initial population of "chromosomes," where each chromosome represents a different subset of financial indicators (factors) from the dataset.
2. For each subset, an OLS regression model is trained to predict the target variable.
3. The "fitness" of each subset is evaluated based on the performance of its corresponding regression model. The primary metrics for fitness were the R-squared (R2) value and the statistical significance of the factors (p-values). A higher R2 indicates a better model fit.
4. The algorithm then evolves the population of factor subsets through processes mimicking natural selection (selection, crossover, and mutation) to give a combination that maximises the regression model with R-squared (R2).

2.3 Portfolio Optimisation (Yash Sharma)

In this project we explore various portfolio optimization techniques applied to a selection of 5 stocks. Portfolio optimization is a crucial aspect of investment management, we try to construct a portfolio of assets that maximizes expected return for a given level of risk, or minimize risk for a given level of expected return. The classic approach, pioneered by Harry Markowitz, forms the foundation, we compare it various other models to address its limitations and incorporate different investment objectives and views.We will evaluate their performance on historical data and then test their effectiveness on unseen data to assess their out-of-sample performance and robustness.

Portfolio Optimization Methods Used:

1. Mean-Variance Optimization (Markowitz): This is the foundational theory of portfolio optimization. It seeks to find portfolios that offer the highest expected return for a defined level of risk (volatility) or the lowest risk for a given expected return. It relies on historical mean returns and the covariance matrix of asset returns.
2. Minimum Variance Portfolio: A specific case of Markowitz optimisation that aims to find the portfolio with the lowest possible volatility, regardless of expected return.
3. Risk Parity Portfolio: This approach focuses on allocating capital such that each asset or risk component contributes equally to the overall portfolio risk. It aims to create a more balanced portfolio in terms of risk contributions, rather than just capital allocation.
4. Conditional Value at Risk (CVaR) Optimisation: This optimisation aims to minimise the expected loss given that the loss exceeds a certain threshold (Value at Risk - VaR). This method is particularly useful for investors concerned about tail risk and potential large losses.
5. Genetic Algorithm Optimisation: This is a heuristic optimisation technique inspired by natural selection. It uses processes like mutation, crossover, and selection to iteratively improve a population of potential portfolio weights, searching for portfolios that maximize a defined fitness function (e.g., Sharpe Ratio). This can be useful for complex problems or when traditional optimisation methods struggle.
6. Robust Optimisation (Ledoit-Wolf Shrinkage): This method addresses the issue of estimation error in the covariance matrix, which can significantly impact the performance of traditional Mean-Variance optimisation, especially with limited historical data. Ledoit-Wolf shrinkage is a technique used here to create a more stable and reliable covariance estimate by shrinking the sample covariance matrix towards a structured estimator. The optimization then uses this robust covariance matrix.

3. Results

3.1 Bayesian Networks

The Bayesian Network models demonstrated effective predictive performance across all five companies, achieving accuracies ranging from73% to 76%. This level of accuracy validates the use of probabilistic models for capturing the intricate dependencies within financial data to forecast stock returns.

Additionally, from the graphs, we can also see how the same variables have different dependencies on each other for all 5 companies.

Key observations from the results include:

Highest Accuracy: The model for Johnson & Johnson (JNJ) achieved the highest overall accuracy at 75.9%.

Best F1-Score: The model for Apple (AAPL) yielded the highest F1-Score of 0.774. The F1-Score, which is the harmonic mean of precision and recall, suggests that the AAPL model provided the best balance between correctly identifying positive returns and avoiding false positives.

Precision and Recall Trade-off: The models for JPMorgan Chase & Co. (JPM) and JNJ demonstrated high precision (0.861 and 0.845, respectively), indicating a low rate of false positives. Conversely, the AAPL model showed a very high recall (0.875), meaning it was highly effective at identifying all actual positive return instances.

In conclusion, the results confirm that Bayesian Networks are a powerful and reliable tool for this financial prediction task. The consistent performance across diverse sectors underscores the model's robustness and its capability to serve as a foundational element in a data-driven investment strategy.

The following are the outputs of the results and graphs for all 5 companies:

```
Company: JPM - Confusion Matrix
[[63 11]
 [32 68]]

Classification Report
              precision    recall  f1-score   support

           0       0.66      0.85      0.75        74
           1       0.86      0.68      0.76       100

    accuracy                           0.75       174
   macro avg       0.76      0.77      0.75       174
weighted avg       0.78      0.75      0.75       174
```

```
Company: JNJ - Confusion Matrix
[[61 13]
 [29 71]]

Classification Report
              precision    recall  f1-score   support

           0       0.68      0.82      0.74        74
           1       0.85      0.71      0.77       100

    accuracy                           0.76       174
   macro avg       0.76      0.77      0.76       174
weighted avg       0.77      0.76      0.76       174
```

```
Company: XOM - Confusion Matrix
[[58 18]
 [29 69]]

Classification Report
              precision    recall  f1-score   support

           0       0.67      0.76      0.71        76
           1       0.79      0.70      0.75        98

    accuracy                           0.73       174
   macro avg       0.73      0.73      0.73       174
weighted avg       0.74      0.73      0.73       174
```

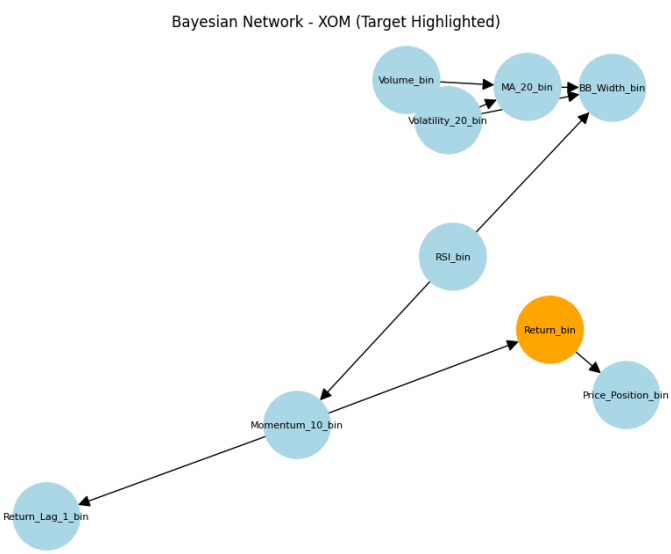
```
Company: AAPL - Confusion Matrix
[[52 34]
 [11 77]]

Classification Report
              precision    recall  f1-score   support

           0       0.83      0.60      0.70        86
           1       0.69      0.88      0.77        88

    accuracy                           0.74       174
   macro avg       0.76      0.74      0.74       174
weighted avg       0.76      0.74      0.74       174
```

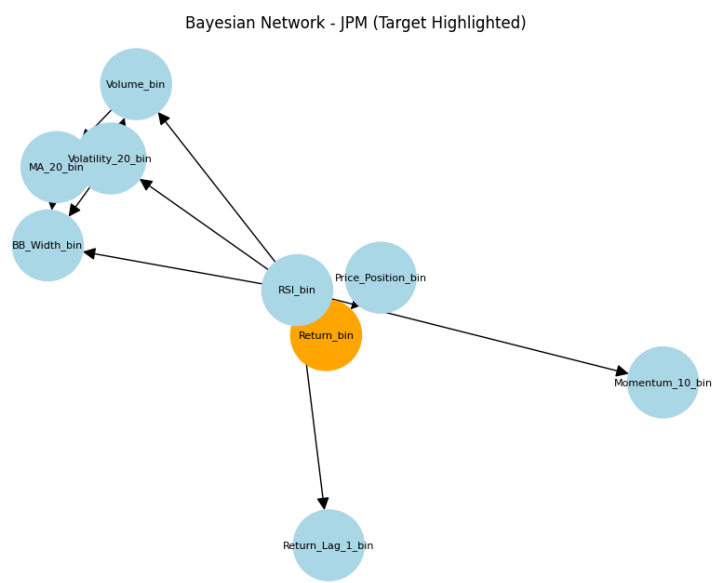
```
Company: PG - Confusion Matrix
[[61 19]
 [24 70]]

Classification Report
              precision    recall  f1-score   support

           0       0.72      0.76      0.74        80
           1       0.79      0.74      0.77        94

    accuracy                           0.75       174
   macro avg       0.75      0.75      0.75       174
weighted avg       0.75      0.75      0.75       174
```

Summary of BN Predictive Performance for All Companies:

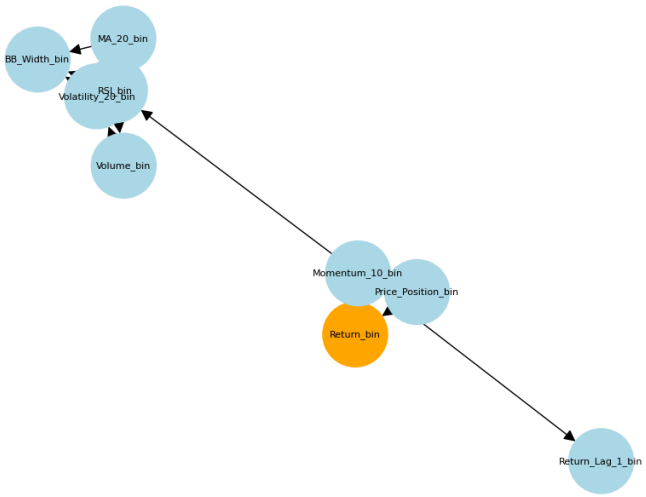
	Company	Accuracy	Precision	Recall	F1-Score
0	PG	0.753	0.787	0.745	0.765
1	AAPL	0.741	0.694	0.875	0.774
2	JNJ	0.759	0.845	0.710	0.772
3	JPM	0.753	0.861	0.680	0.760
4	XOM	0.730	0.793	0.704	0.746

graphs:

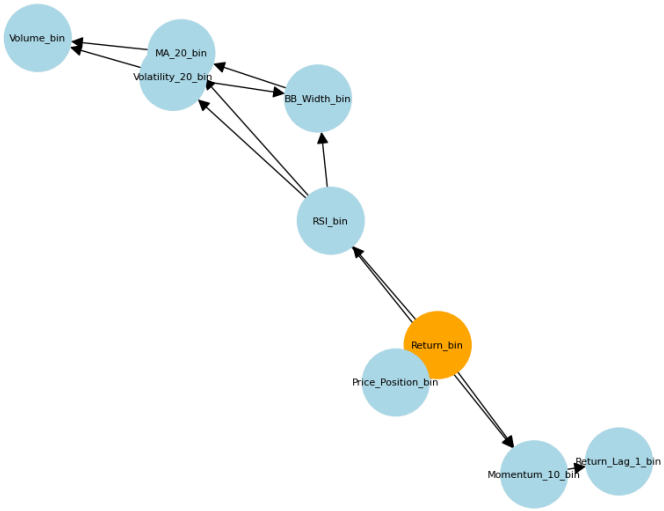




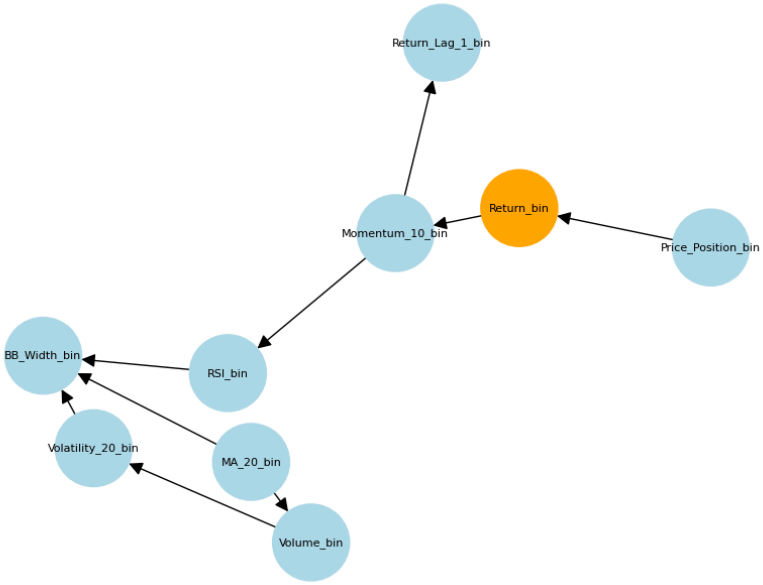
Bayesian Network - JNJ (Target Highlighted)



Bayesian Network - AAPL (Target Highlighted)



Bayesian Network - PG (Target Highlighted)



3.2 Genetic Algorithm with Regression

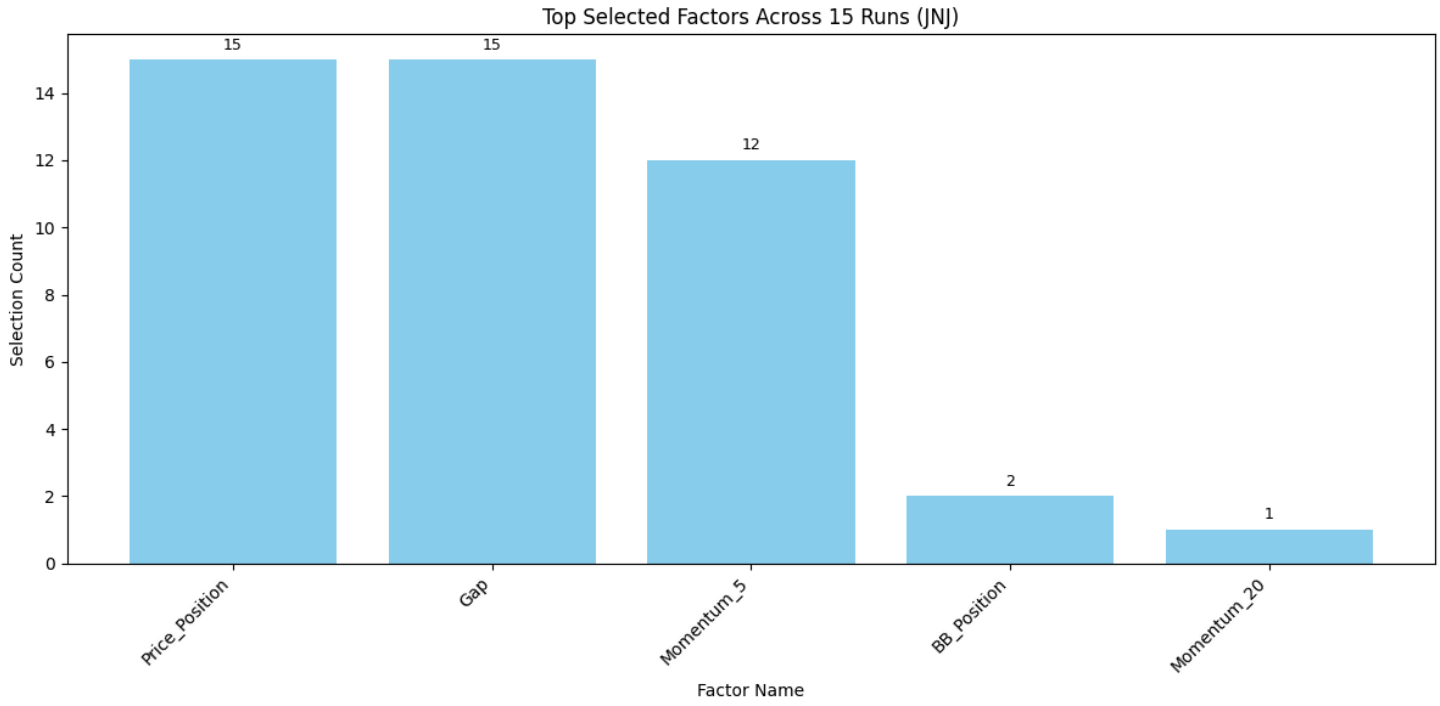
The application of the Genetic Algorithm with Regression proved highly effective in identifying significant sets of predictors for each of the five companies. The OLS regression results for the factor combinations chosen by the GA showed strong explanatory power, with high R2 values across the board:

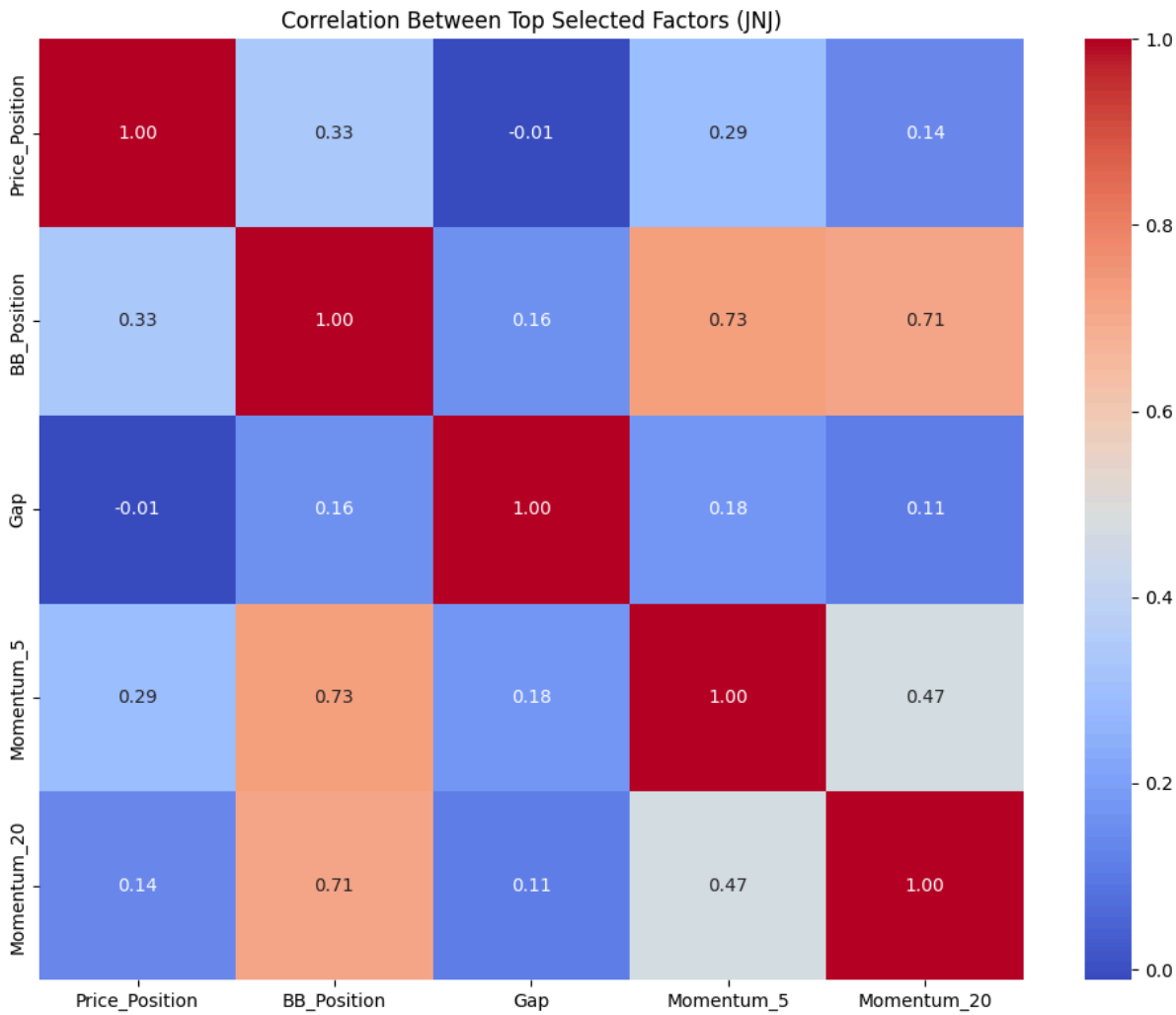
- **JPM:** R2=0.761
- **AAPL:** R2=0.717
- **JNJ:** R2=0.686
- **XOM:** R2=0.655
- **PG:** R2=0.636

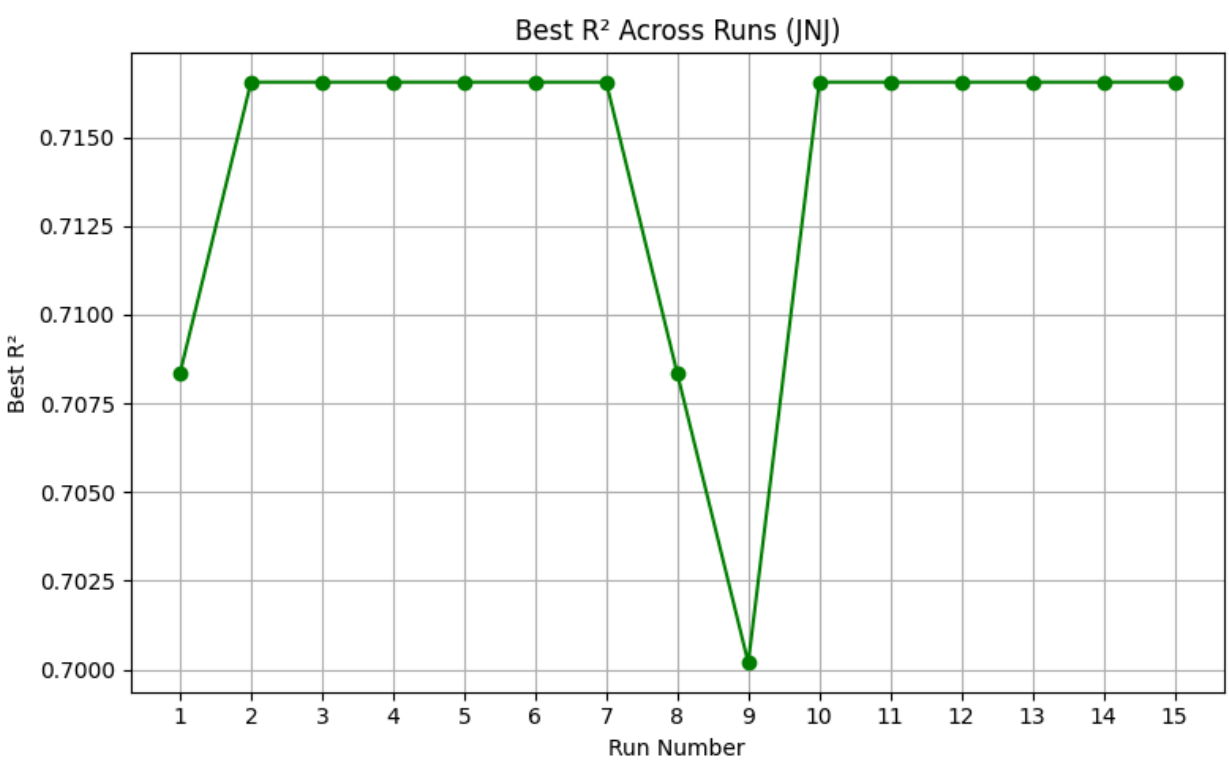
These results indicate that the selected factors account for a significant portion of the variance in stock returns for each respective company.

The following are the graphs showing the number of occurrences of factors through multiple runs, correlation heatmap and R2 values throughout the runs for each company:

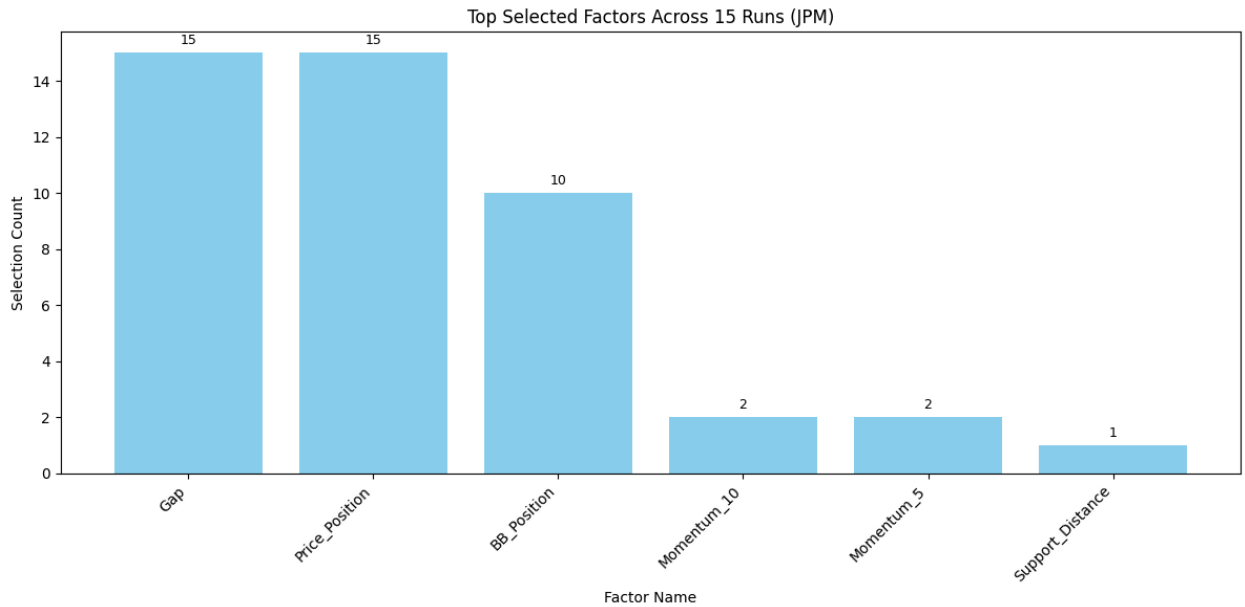
Johnson & Johnson (JNJ) :

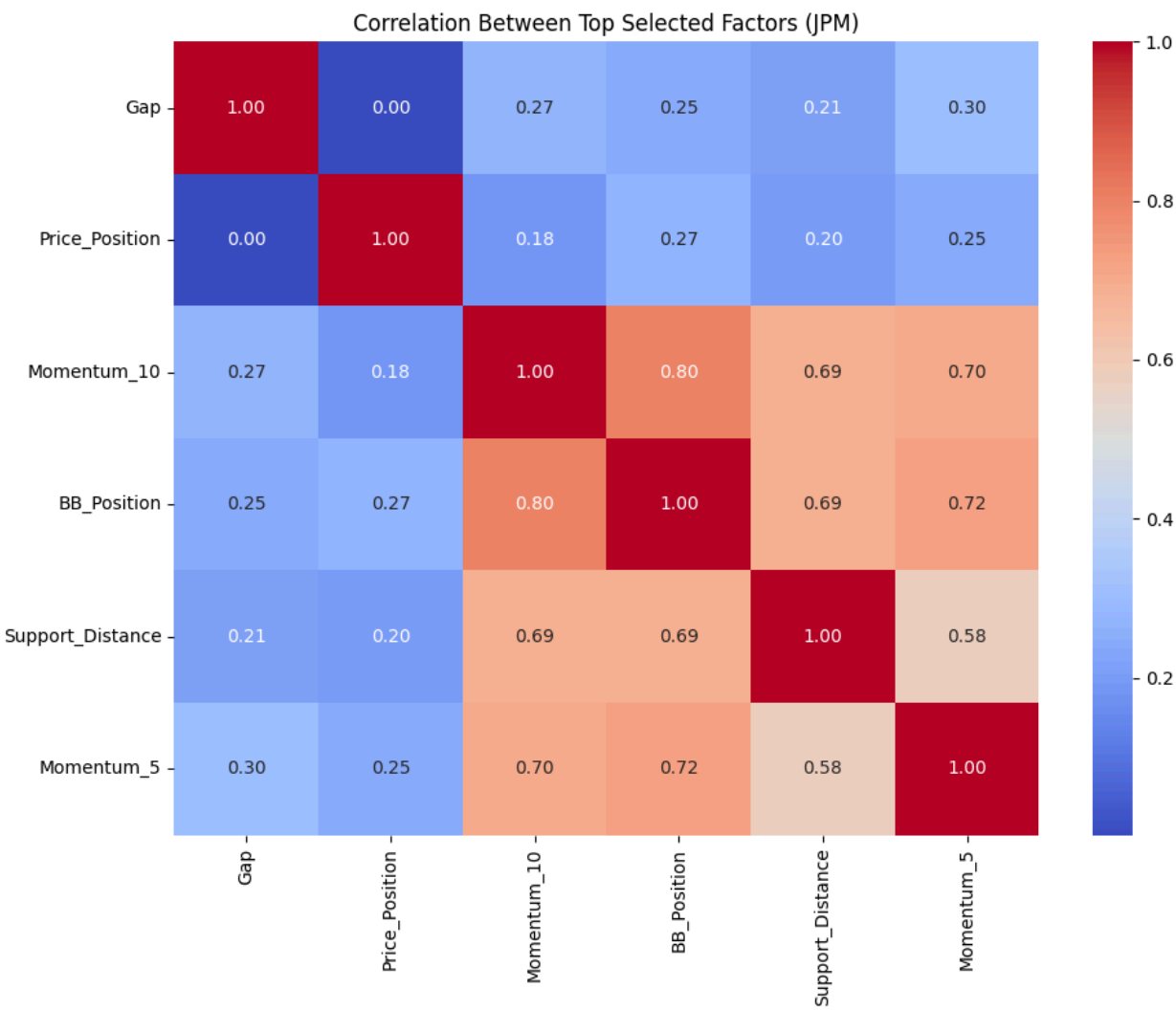


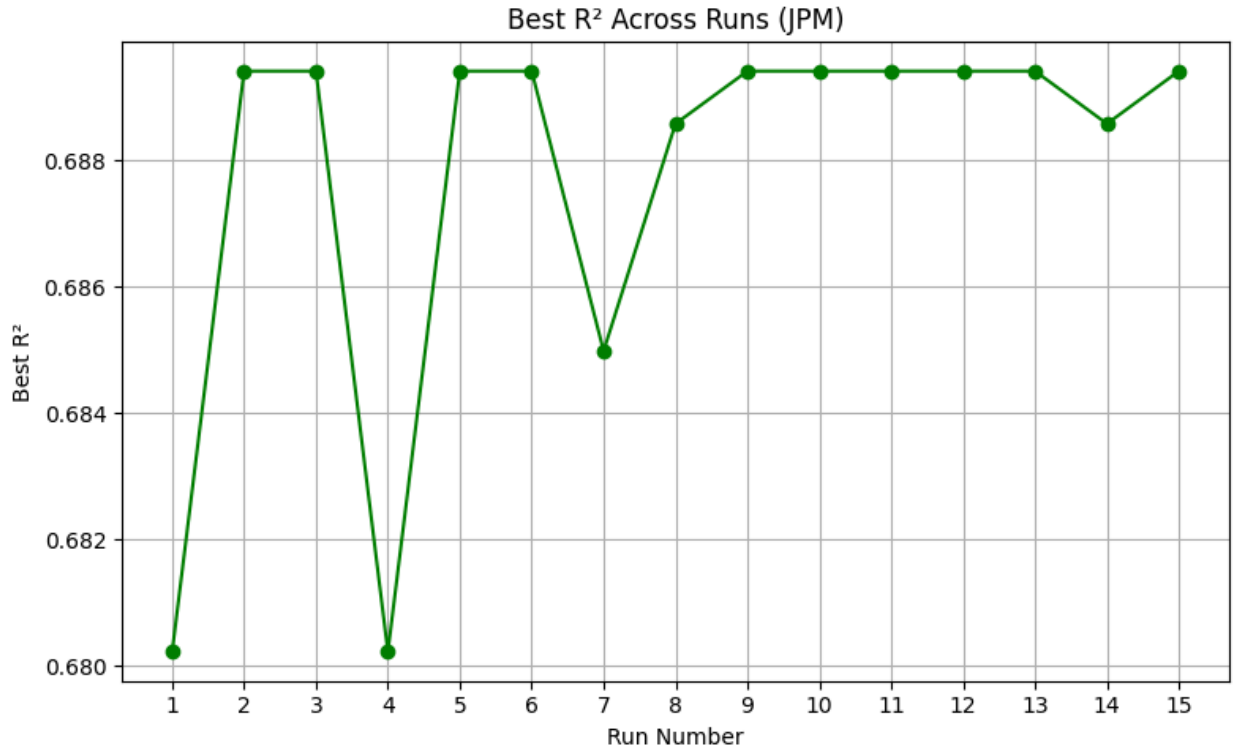

Correlation Between Top Selected Factors (JNJ)
Price_Position
BB_Position
Gap
Momentum_5
Momentum_20
1.00
0.33
-0.01
0.29
0.14
0.33
1.00
0.16
0.73
0.71
-0.01
0.16
1.00
0.18
0.11
0.29
0.73
0.18
1.00
0.47
0.14
0.71
0.11
0.47
1.00
1.0
0.8
0.6
0.4
0.2
0.0


Best R² Across Runs (JNJ)
Best R²
0.7150
0.7125
0.7100
0.7075
0.7050
0.7025
0.7000
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
Run Number

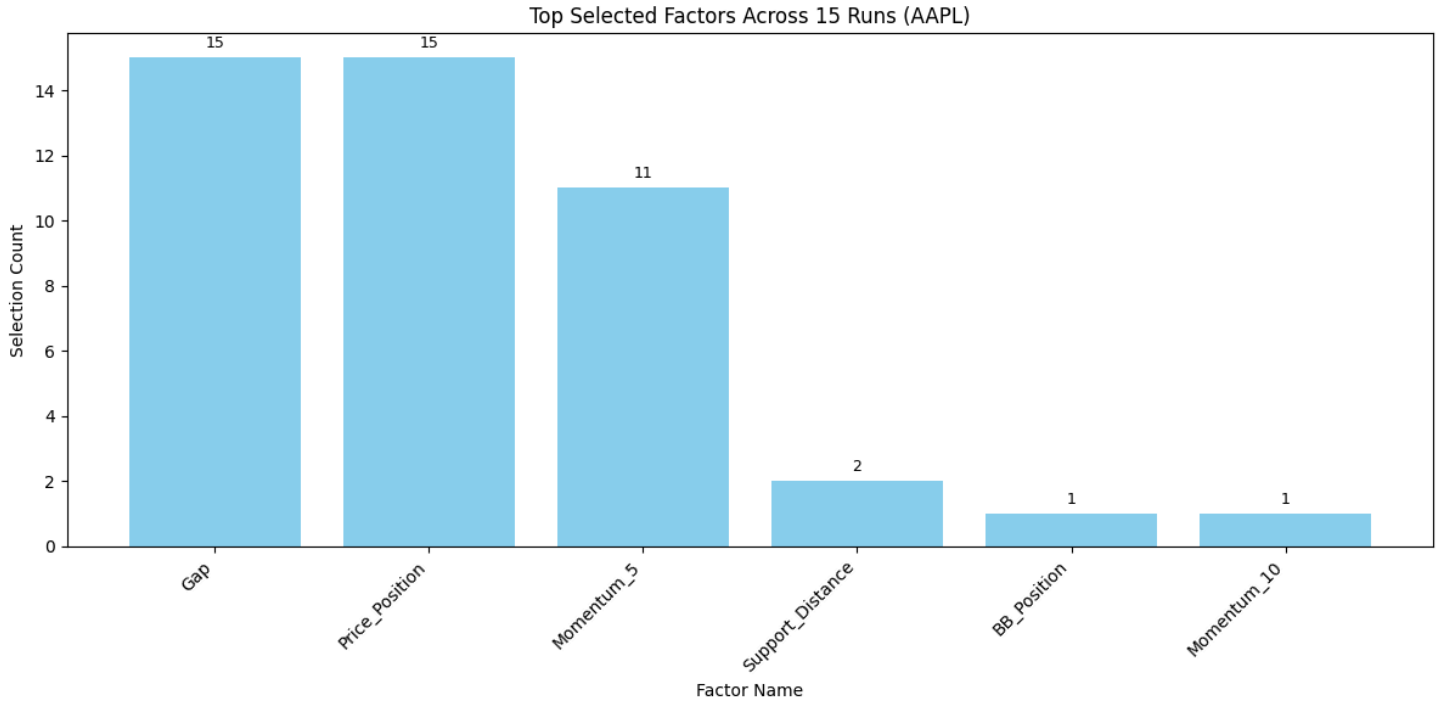
JPMorgan Chase & Co. (JPM)

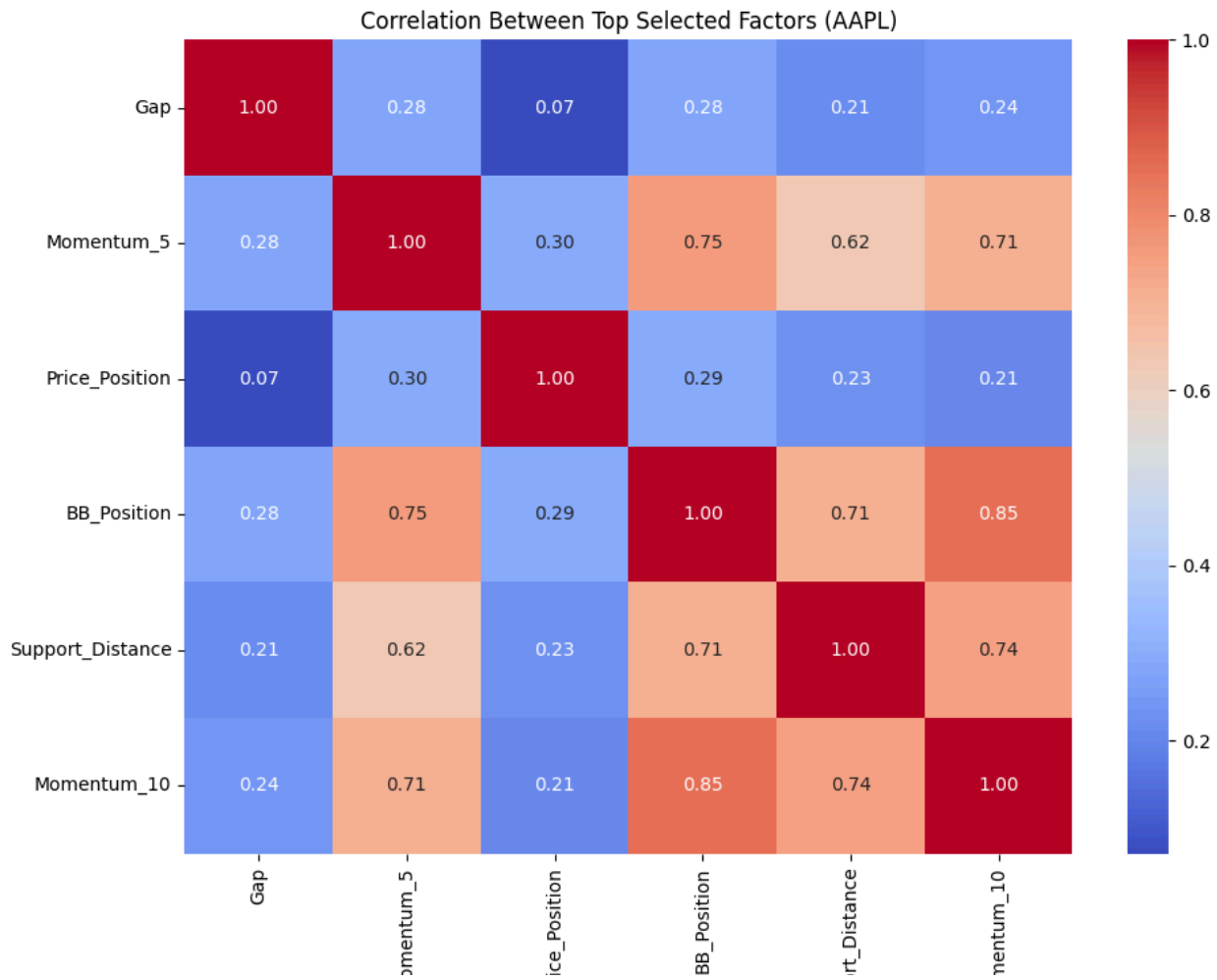


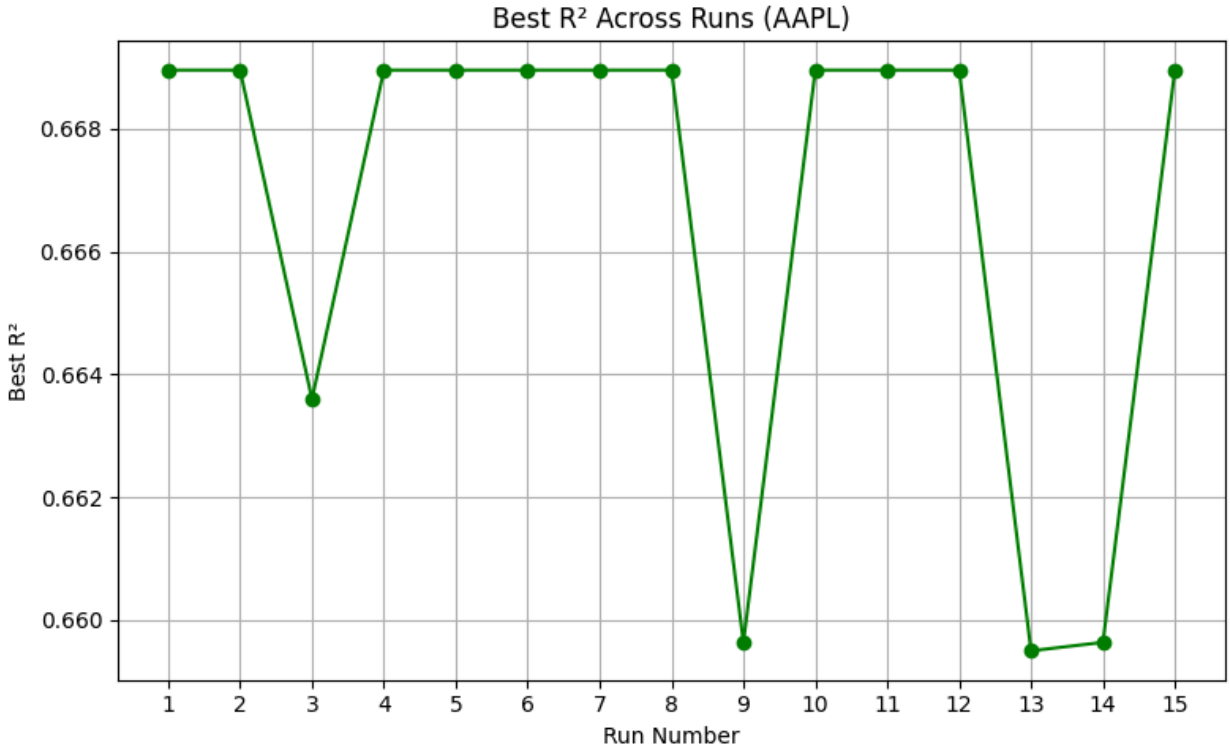



Best R² Across Runs (JPM)
Best R²
0.680
0.682
0.684
0.686
0.688
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
Run Number

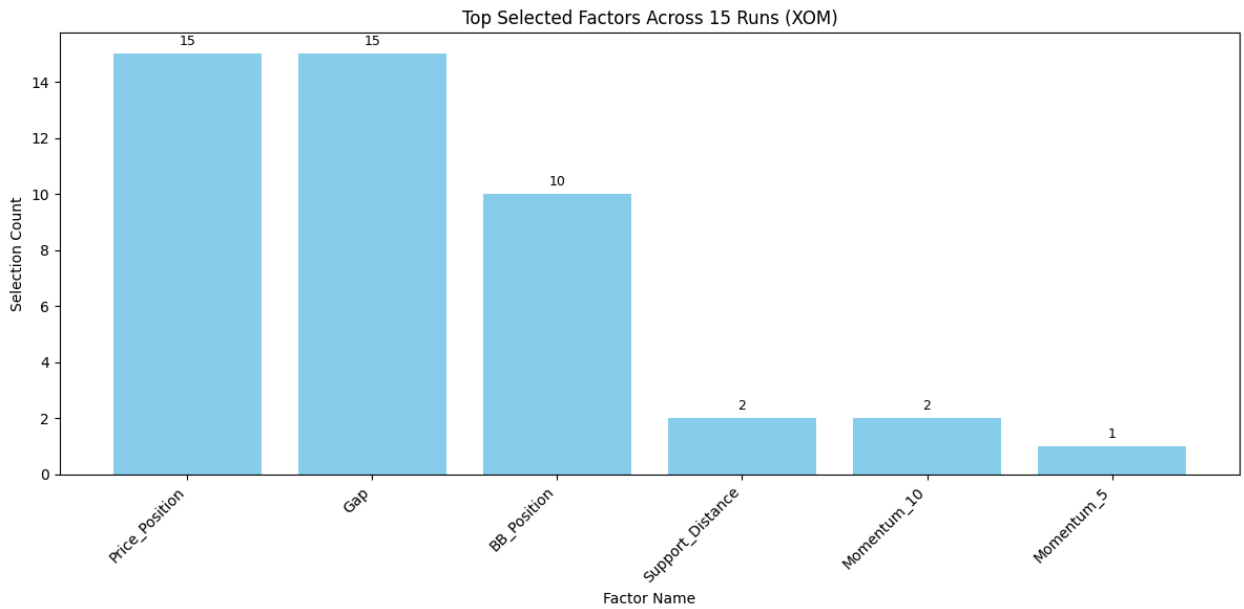
Apple (AAPL) :

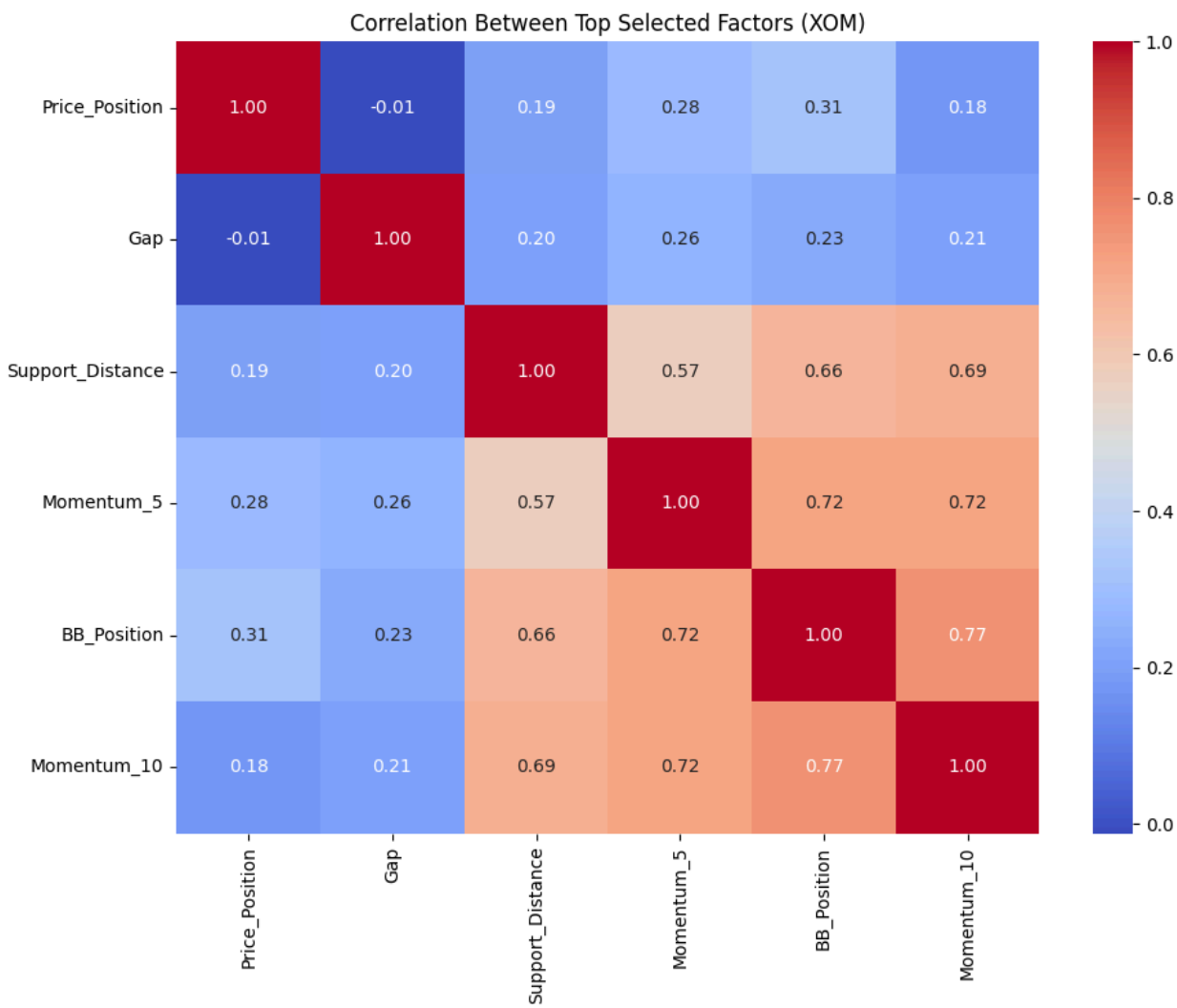

Top Selected Factors Across 15 Runs (AAPL)
Selection Count
0
2
4
6
8
10
12
14
15
15
11
2
1
1
Gap
Price_Position
Momentum_5
Support_Distance
BB_Position
Momentum_10
Factor Name

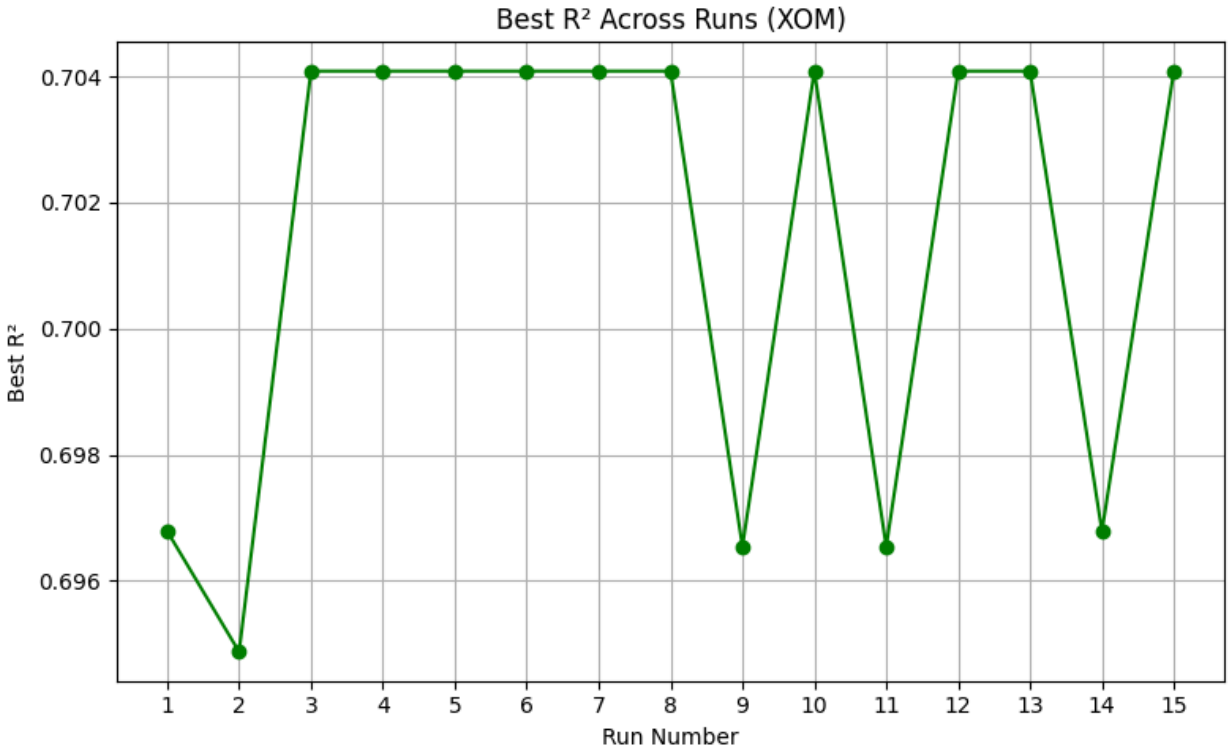

Correlation Between Top Selected Factors (AAPL)
Gap
Momentum_5
Price_Position
BB_Position
Support_Distance
Momentum_10
1.00 0.28 0.07 0.28 0.21 0.24
0.28 1.00 0.30 0.75 0.62 0.71
0.07 0.30 1.00 0.29 0.23 0.21
0.28 0.75 0.29 1.00 0.71 0.85
0.21 0.62 0.23 0.71 1.00 0.74
0.24 0.71 0.21 0.85 0.74 1.00
1.0
0.8
0.6
0.4
0.2


Best R² Across Runs (AAPL)
Best R²
0.668
0.666
0.664
0.662
0.660
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Run Number

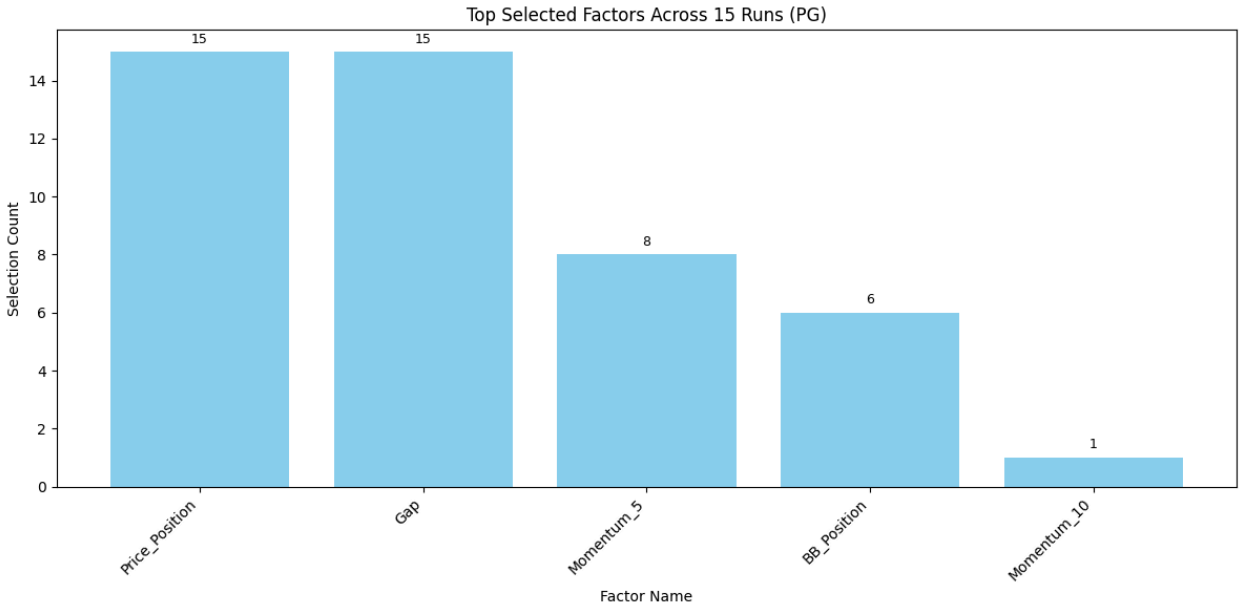
Exxon Mobil Corporation (XOM) :

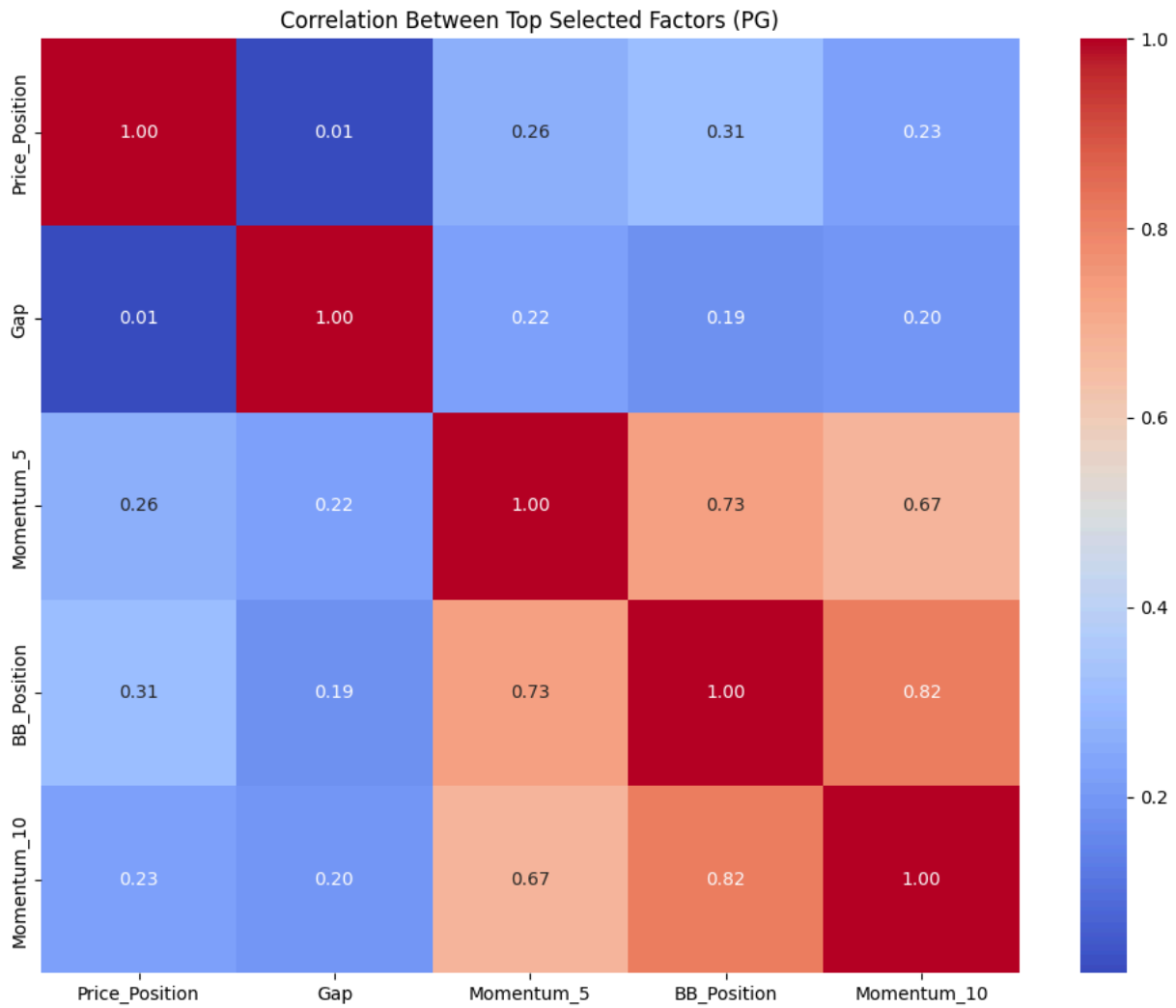


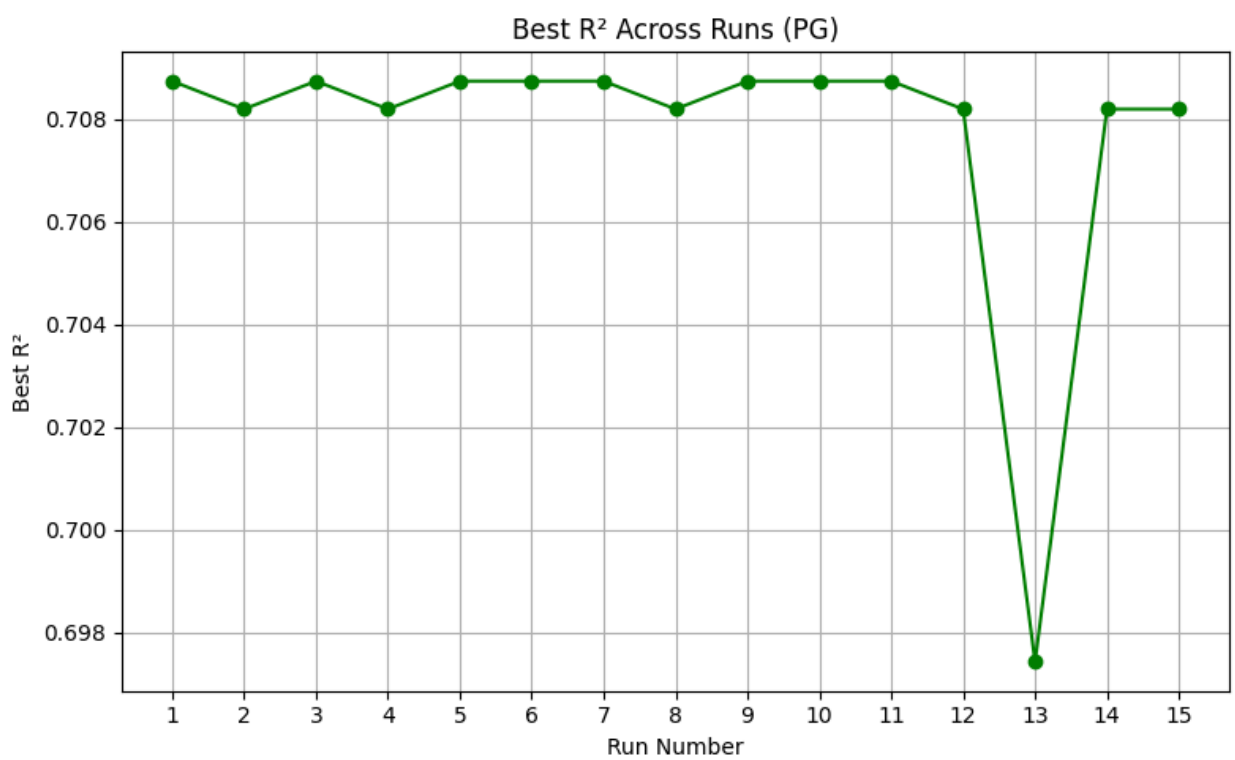




Procter & Gamble (PG) :




Correlation Between Top Selected Factors (PG)
Price_Position
Gap
Momentum_5
BB_Position
Momentum_10
1.00 0.01 0.26 0.31 0.23
0.01 1.00 0.22 0.19 0.20
0.26 0.22 1.00 0.73 0.67
0.31 0.19 0.73 1.00 0.82
0.23 0.20 0.67 0.82 1.00
1.0
0.8
0.6
0.4
0.2


Best R² Across Runs (PG)
Best R²
0.708
0.706
0.704
0.702
0.700
0.698
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Run Number

3.3 Portfolio Optimisation

The analysis was conducted in two phases: training on historical data and testing on unseen data.

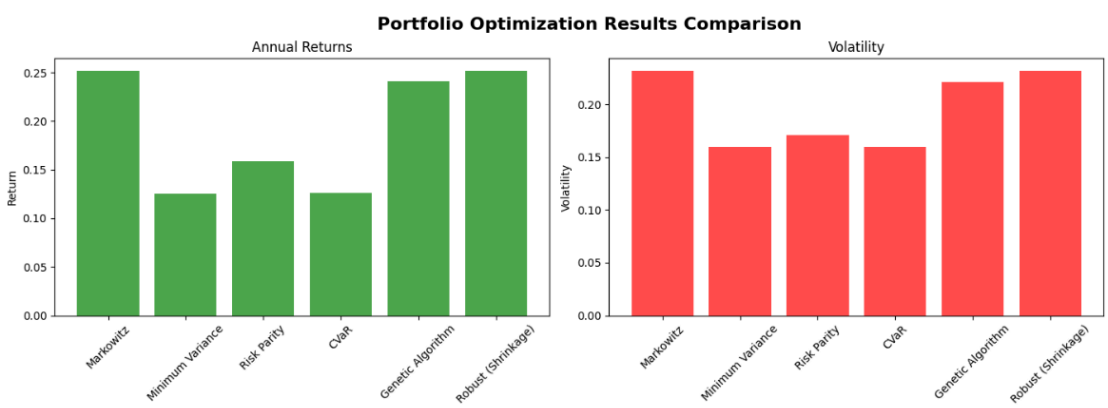
3.3.1 Phase 1: Training on Historical Data

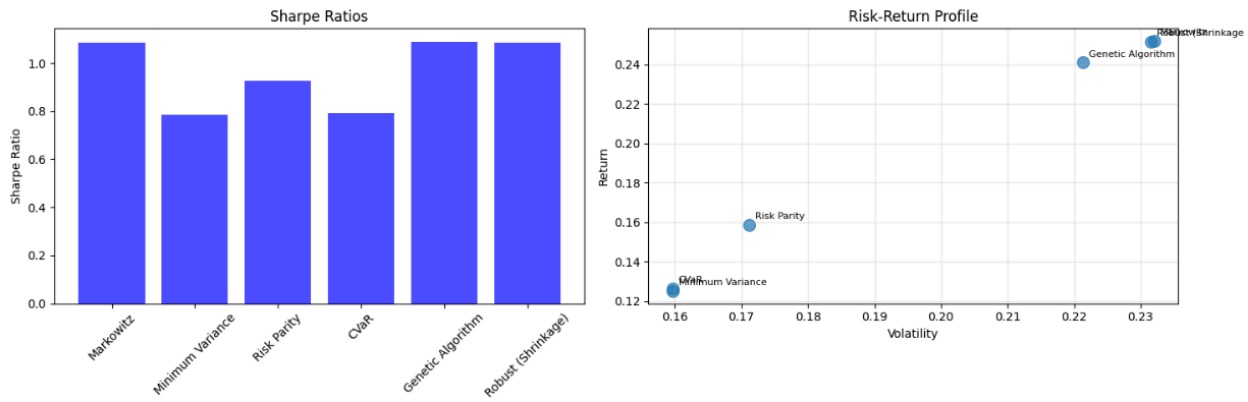
The portfolio optimisation methods were first applied to historical price data from 2016-01-01 to 2023-12-31. The table below summarises the performance metrics for each method

Method	Annual Return	Volatility	Sharpe Ratio
Markowitz	0.252	0.232	1.086
Minimum Variance	0.125	0.160	0.784
Risk Parity	0.158	0.171	0.926
CVaR	0.126	0.160	0.790
Genetic Algorithm	0.241	0.221	1.089
Robust (Shrinkage)	0.251	0.232	1.086

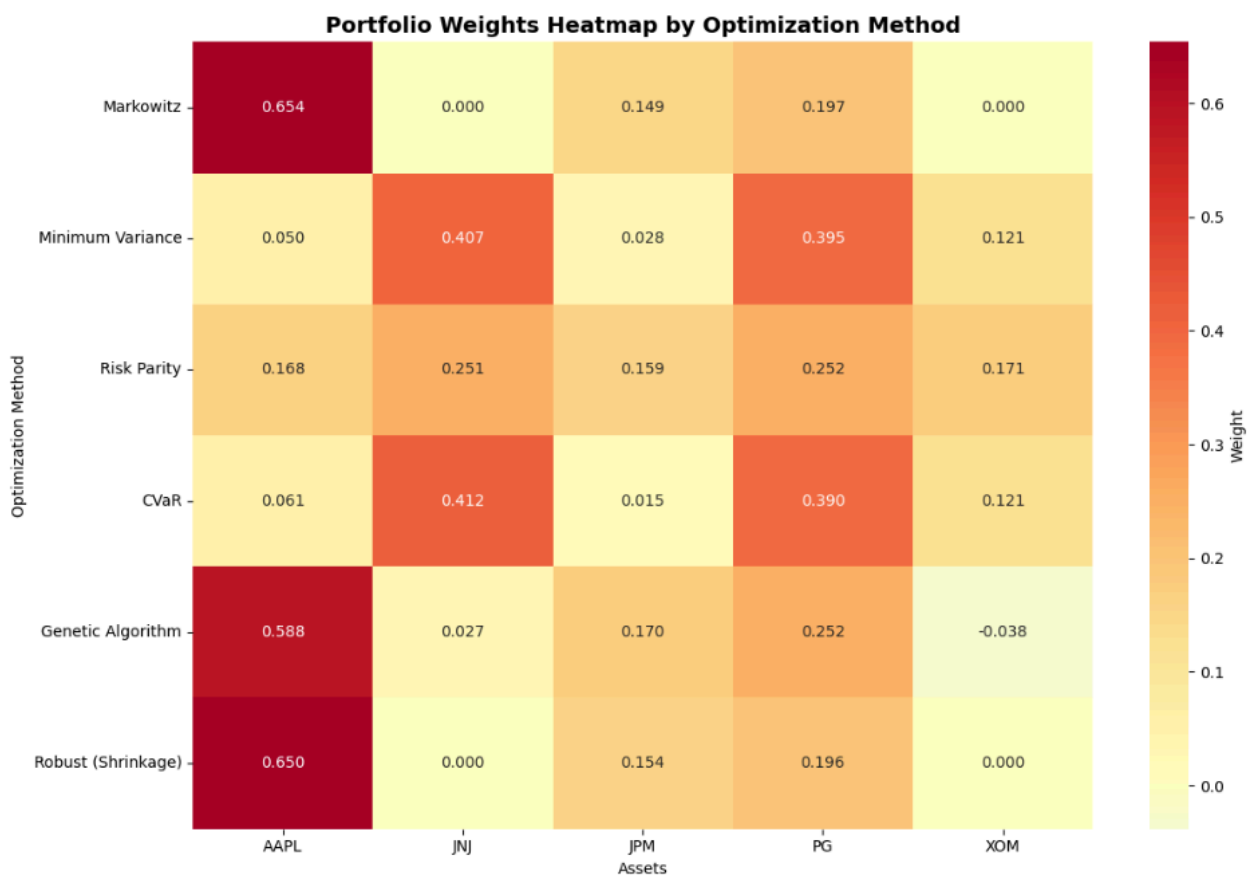
- The Genetic Algorithm and Markowitz methods achieved the highest Sharpe Ratios on the training data, indicating a better risk-adjusted return.
- The Minimum Variance and CVaR methods resulted in lower volatility portfolios, as expected, but also lower returns.
- Robust Optimisation performed very similarly to the standard Markowitz approach on this dataset.

The following plots visualise these training phase results:

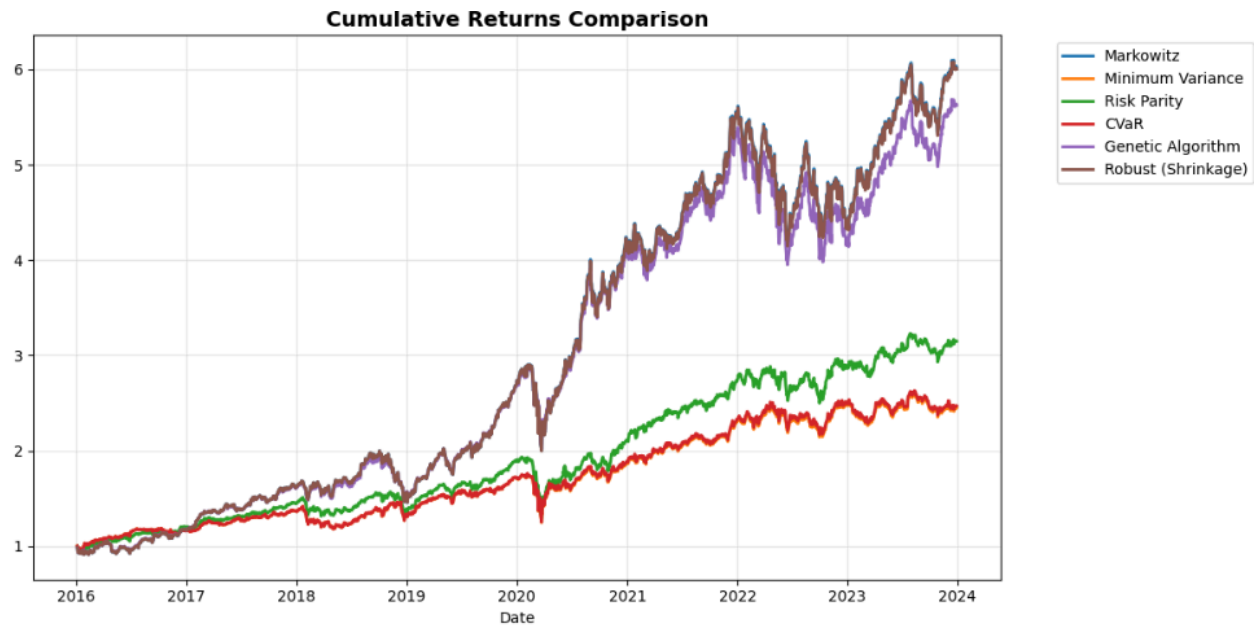




Portfolio Optimisation Results Comparison: This plot shows a bar chart comparing the annual returns, volatility, and Sharpe ratios of the different optimisation methods on the training data. It also includes a risk-return scatter plot.



Portfolio Weights Heatmap by Optimisation Method: This heatmap visualises the allocation of weights to each asset for every optimisation method. It highlights how different strategies distribute capital across the assets.



Cumulative Returns Comparison: This plot shows the cumulative growth of an investment in each portfolio over the training period. It helps in understanding the historical performance trajectory of each strategy.

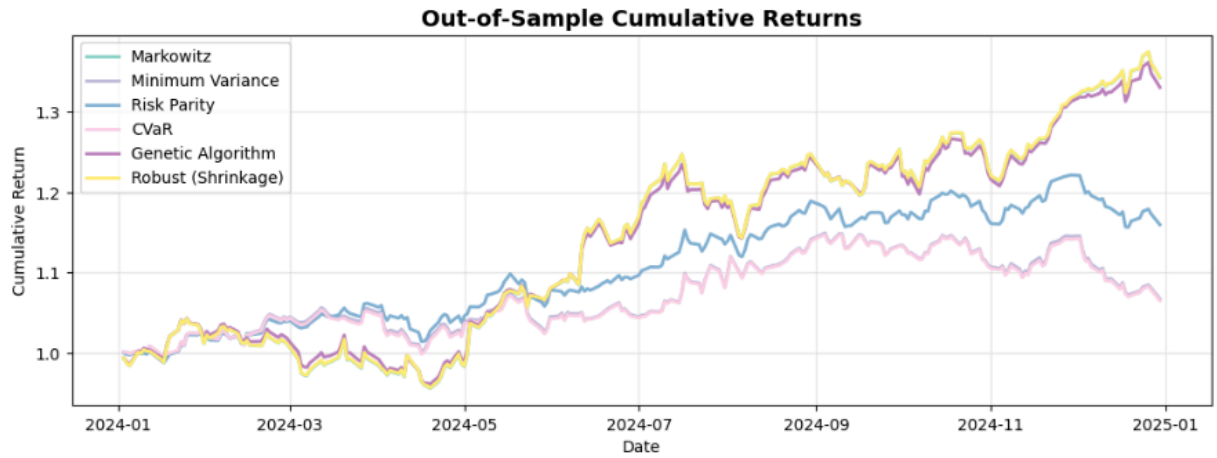
3.3.2 Phase 2: Testing on Unseen Data

To evaluate the out-of-sample performance and robustness of the optimised portfolios, they were tested on unseen price data from 2023-12-31 to 2024-12-31. The table below presents the key performance metrics on this unseen data:

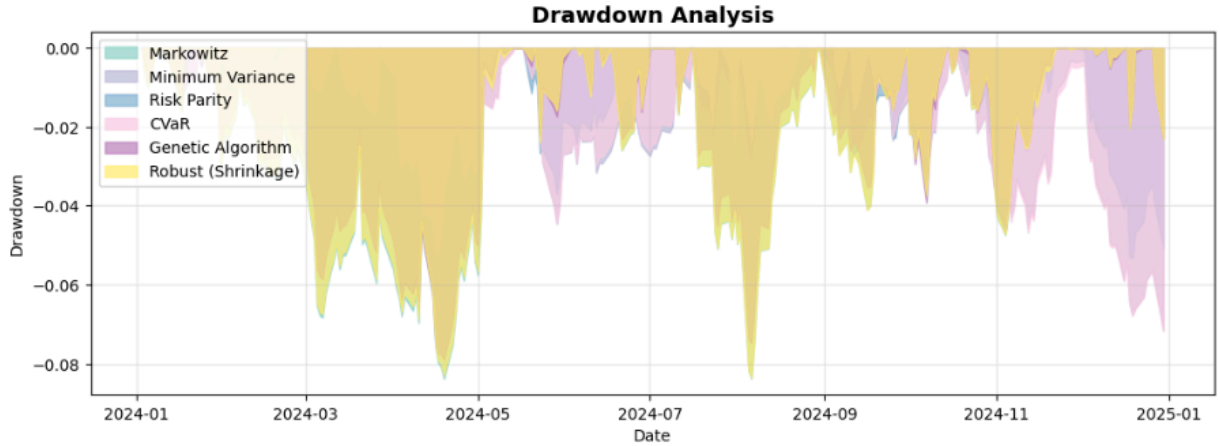
Strategy	Total Return	Annualized Return	Volatility	Sharpe Ratio	Sortino Ratio	Maximum Drawdown
Markowitz	0.3430	0.3462	0.1534	2.1265	3.3162	-0.0838
Minimum Variance	0.0671	0.0676	0.1027	0.4635	0.7188	-0.0714
Risk Parity	0.1596	0.1609	0.0940	1.4991	2.1895	-0.0533
CVaR	0.0650	0.0656	0.1025	0.4442	0.6878	-0.0717
Genetic Algorithm	0.3308	0.3339	0.1424	2.2036	3.4193	-0.0790
Robust (Shrinkage)	0.3435	0.3466	0.1527	2.1389	3.3338	3.3338

- On unseen data, the Genetic Algorithm and Robust (Shrinkage) methods demonstrated the highest Sharpe and Sortino Ratios, indicating superior risk-adjusted performance.
- Risk Parity showed the lowest maximum drawdown, suggesting better performance during adverse market conditions, despite having a lower Sharpe Ratio than the top performers.
- The results on unseen data highlight the importance of out-of-sample testing, as the relative performance of strategies can differ from the training period.

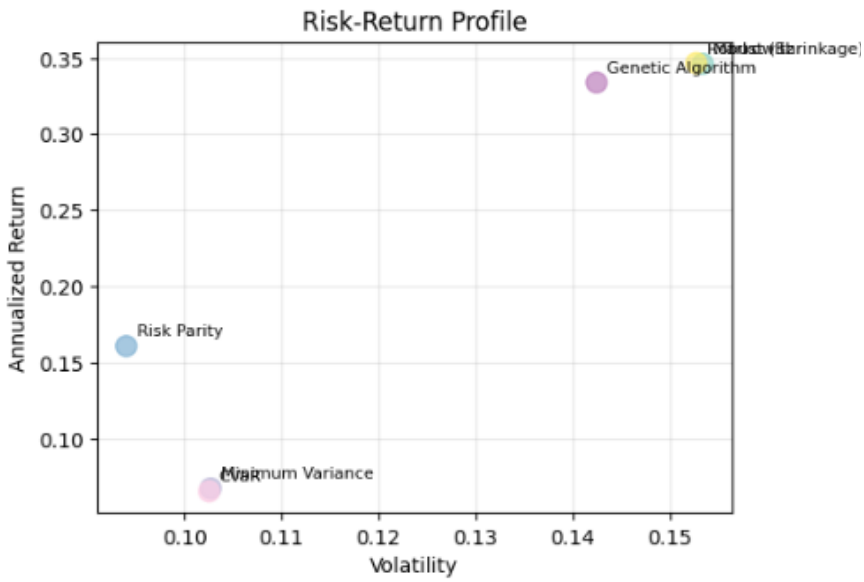
The following plots provide a comprehensive analysis of the out-of-sample test results:



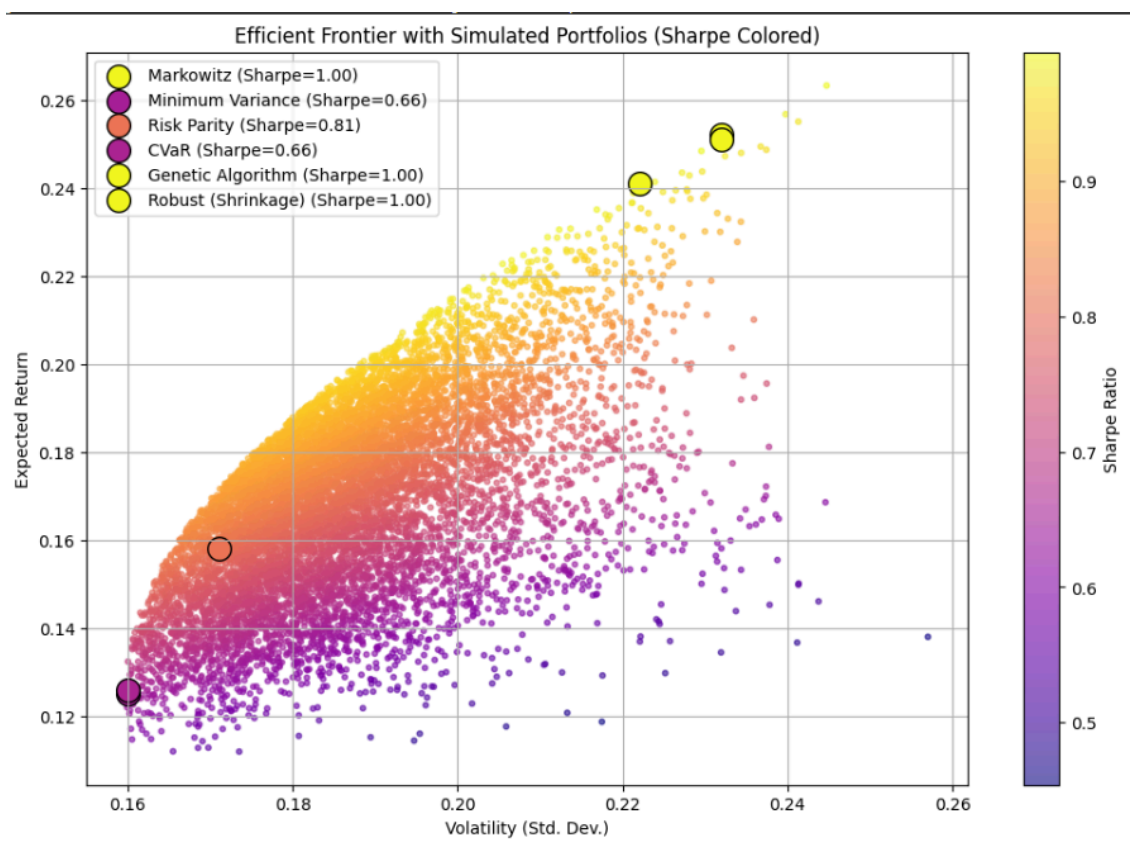
Out-of-Sample Cumulative Returns: This plot displays the cumulative returns of each strategy on the unseen data, illustrating their performance over the test period.



Drawdown Analysis: This chart shows the drawdowns experienced by each portfolio over the unseen data period, highlighting their resilience during market downturns.



Risk-Return Profile: A scatter plot showing the annualised return vs. volatility for each strategy on the unseen data.



Efficient Frontier : this graph is useful to depict the efficient frontier for the portfolios.

4. Conclusion and Future Work

4.1 Summary of Findings

This project successfully implemented an integrated data science framework to address the challenges of modern financial market analysis, which cannot be done effectively with traditional methods. All core objectives were successfully met:

1. Predictive Modelling: The objective to predict short-term stock returns was achieved on the test data. The Bayesian Network models demonstrated significant capability in capturing the probabilistic dependencies within the dataset, giving predictive accuracies ranging between 73% and 76% across the five diverse companies. This confirms that advanced probabilistic models are a powerful tool for short-term stock return forecasting.
2. Feature Selection: The project successfully identified the most influential factors of stock performance. The combination of the Genetic Algorithm with a regression proved highly effective in finding the key factors. The consistent selection of these factors offers concrete insight into the market that drives price movement.
3. Portfolio Optimisation: The final objective of constructing investment portfolios was validated through extensive backtesting. Analysis of multiple optimisation strategies revealed that modern computational methods provide an edge over traditional methods. The optimization portfolios delivered better results on the testing data, demonstrating an ability to strategically balance risk and maximise the Sharpe Ratio.

4.2 Project Significance and Impact

The contribution of this project is the successful demonstration that a comprehensive data science pipeline from feature engineering and predictive modelling to portfolio optimisation can provide a significant, quantitative edge against the volatility of financial markets. The results validate the approach of not only predicting returns but also optimising them to achieve a better risk-return ratio.

4.3 Limitations and Future Research

While the project demonstrated strong predictive performance, certain limitations must be acknowledged. The models used exclusively quantitative data; they did not incorporate unstructured and qualitative data, such as news sentiment or social media analysis, which can be critical drivers of market fluctuations.

Future research should focus on three key areas:

1. Deep Learning Integration: Exploring more complex, non-linear models, such as Long Short-Term Memory (LSTM) networks, to potentially capture longer-term dependencies.
2. Sentiment Analysis: Using the current dataset with Natural Language Processing (NLP) outputs from financial news feeds to incorporate qualitative market sentiment.

3. Dynamic Portfolio Rebalancing: Developing a system for real-time portfolio rebalancing based on the model's continuously generated predictions, thus moving toward a more practical and deployable trading strategy.